

ALAN GOMEZ

Universidad del Valle de Guatemala (UVG), Guatemala City

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📍 Zone 16, Guatemala

in LinkedIn

🐙 Github

☁️ AWS Portfolio



EXPERIENCE

Software Development Intern – Value-Added Services (VAS)

Claro Guatemala

📅 Feb 2026 – May 2026

📍 Guatemala City, Guatemala

- Developed and deployed an internal Python and Flask platform that processed Huawei UGW configurations and generated network topology diagrams using Graphviz, reducing manual analysis time.
- Built a secure text-processing web application with Argon2id authentication, user administration, session management, and activity logging.
- Researched and tested automated messaging solutions using the Meta WhatsApp Business API and Twilio.
- Containerized and deployed applications with Docker for use in a production environment.

D-Hive Assistant

University of the Valley of Guatemala

📅 Jan 2024 – Nov 2024

📍 Guatemala City, Guatemala

- Conducted training sessions and safety workshops for university students.
- Maintained 3D printers to ensure optimal operation and minimize downtime.
- Designed components using laser cutters for various projects.

TECHNICAL SKILLS

- **Programming:** Python, C, C++, JavaScript Basic, HTML Basic, CSS Basic, Verilog.
- **Frameworks & Tools:** Flask, Docker, Git/GitHub CLI, CMake, Anaconda, uv, virtual environments, and LaTeX.
- **Operating Systems:** Linux, command-line tools, and Bash scripting.
- **Networking & APIs:** Routing and switching fundamentals, Huawei UGW configuration analysis.

EDUCATION

Bachelor in Electronics Engineering

University of the Valley of Guatemala

📅 Jan 2022 – Present

HOBBIES

- C++/Rust Programming, and Cloud.

PERSONAL SKILLS

- Independent and self-motivated, with a proactive approach to problem-solving.
- Initiative-driven, capable of identifying and implementing solutions efficiently.
- Intermediate English proficiency: strong reading skills, particularly with technical documentation; moderate listening comprehension; able to engage in casual conversations with developing fluency.

PROJECTS

Secure Password Manager

- Developed a Flask-based password manager for securely generating, encrypting, organizing, and managing user credentials.
- Protected sensitive data using PyNaCl Secret-Box symmetric encryption and authenticated user sessions.
- Implemented cryptographically secure password generation, environment-based secrets configuration, and structured CSV/JSON storage.
- Source code: GitHub

Secure Text Processing Platform

- Developed an internal Flask application for processing and converting structured data lists.
- Implemented Argon2id password hashing, secure authentication, session expiration, user administration, and activity logging.
- Containerized and deployed the application with Docker to provide controlled and auditable access in a production environment.

Huawei UGW Configuration & Topology Analyzer

- Developed a Python and Flask platform for analyzing Huawei UGW network configurations.
- Generated standardized network topology diagrams with Graphviz, reducing manual analysis and improving technical documentation.